

Early Warning Indicator

CIC International Seminar

Niko Plennis and Robin Huller – June 2022

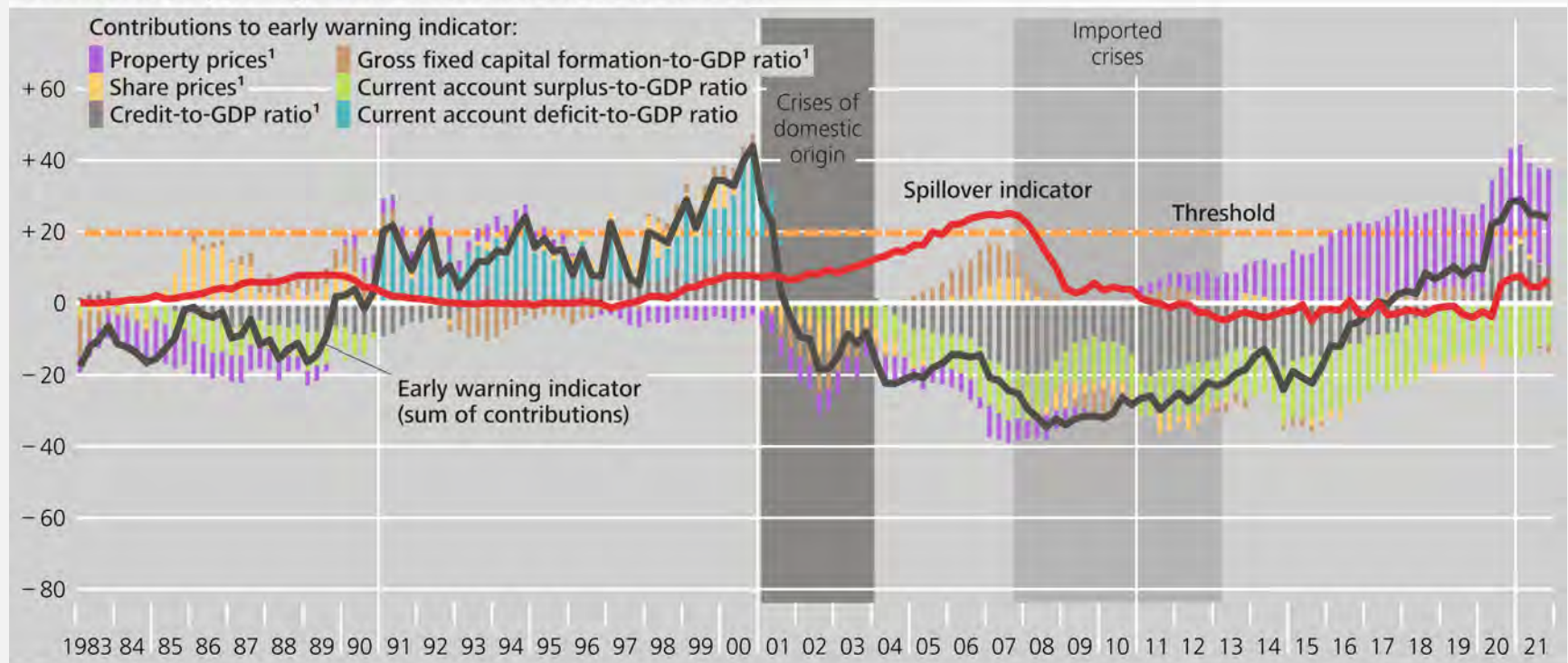
The views expressed in the following slides are mine and do not necessarily reflect the views of Deutsche Bundesbank.

- Introduction
- Model setup
- Evaluation of the EWI
- Variable selection of the EWI
 - Literature/ Meta analysis
 - Eventplots
 - EWI Model evaluation
- Counterfactual EWI analysis in light of the Corona pandemic

The Early Warning Indicator (EWI) for Germany shows a rising trend since 2015 driven by property prices and the credit-to-GDP gap.

Early warning indicator and spillover indicator for Germany

Normalisation: early warning indicator in the United States in Q1 2006 = 100



Sources: BIS, Bloomberg, ECB, Eurostat, IMF, OECD, Refinitiv, Bundesbank statistics and Bundesbank calculations. ¹ Cyclical deviation from the long-term trend, based on the Hodrick-Prescott (HP) filter.

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Introduction 1/2: What are Early Warning Models and how do we use them?

Descriptive indicators

Composite indicators based on econometric techniques

Stress tests and model-based analyses

Expert judgment



Overall assessment of vulnerabilities and resilience

Introduction 2/2: What are Early Warning Models and how do we use them?

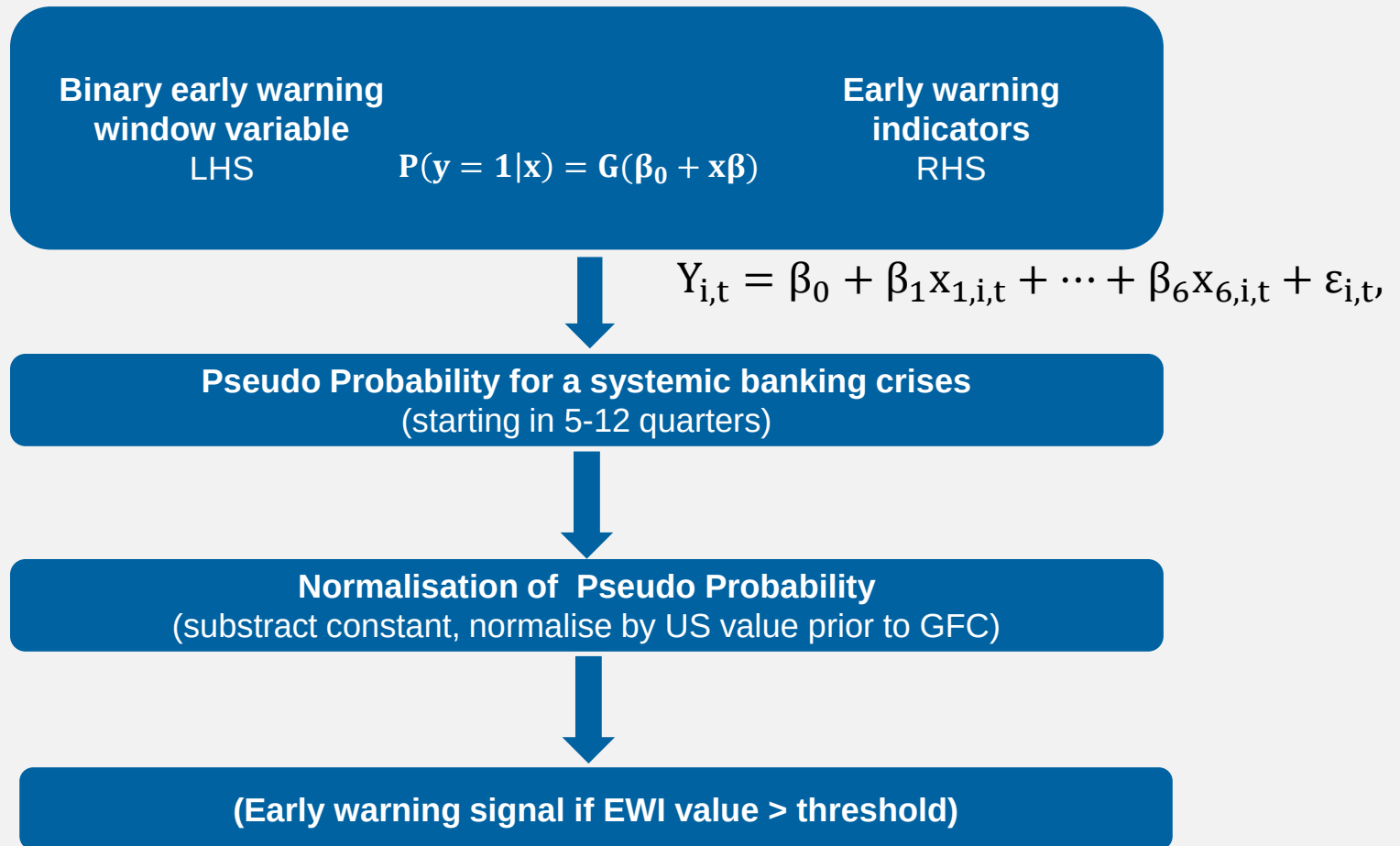
- Aim: Monitor to which extent vulnerabilities are (endogenously) building up in the financial system based on historical data patterns.
 - Pre-requisite: Identify macrofinancial variables which have predictive power for (historical) financial crises (“Early Warning Indicators”).
 - Weakness: New types of crises are difficult to predict based on historical data patterns.
- Early warning models form part of the macroprudential surveillance tool-kit and should not be used as single tool for financial stability assessment.

Question:

Which indicators do you use to monitor a potential build-up of vulnerabilities in the financial system?

Model setup

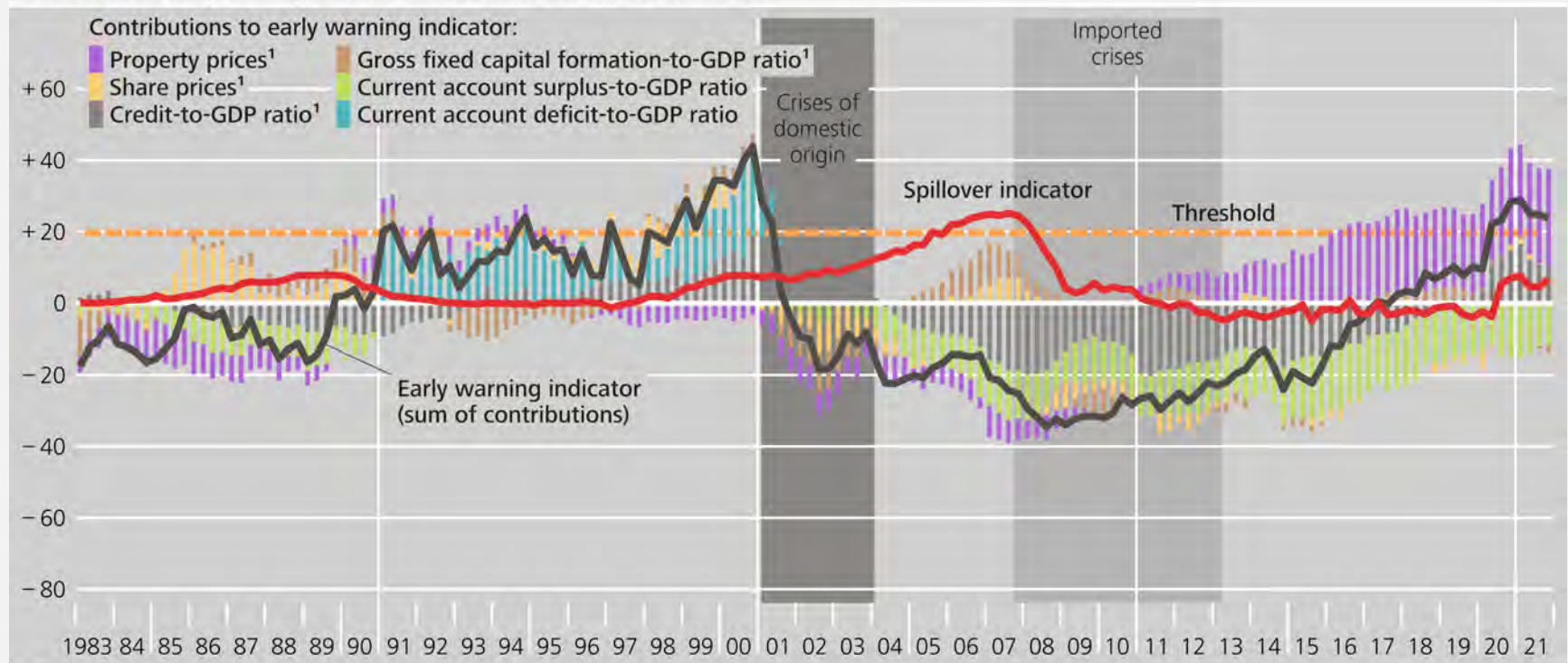
Model setup 1/4: General setup



The Early Warning Indicator (EWI) for Germany shows a rising trend since 2015 driven by property prices and the credit-to-GDP gap.

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Model setup 2/4: Dependent variable

Dependent variable: Early Warning variable for crises (binary)

- Based on **crises database** established by the European System of Central Banks (ESCB) / European Systemic Risk Board (ESRB)
 - Contains crises dates for all **EU countries** and **Norway** from 1970-2021 (Lo Duca et al., 2017)
 - Enhanced with crises dates for **US and Japan** from the dataset by Laeven and Valencia (2012)
- Consideration of **systemic banking crises** with **at least partially domestic** origins
- Estimation sample: 19 crises for European countries (of which 11 crises before 2008), 3 crises added for USA and Japan.
- **Only tranquil and early warning periods enter estimation sample.**

Model setup 3/4: Explanatory variables

Explanatory variables: Early warning indicators

Time period: 1971 Q1 – 2021 Q4 (unbalanced panel)

Data sources: Bundesbank, BIS, ECB, Eurostat, IMF, OECD

Channels:

- **Asset Prices:** Property prices, equity prices
- **Credit:** Credit-to-GDP
- **Macro:** Gross fixed capital formation to GDP
- **External / global:** Current-account-to-GDP
- **Other channels:** Restricted by data availability (e.g. banking data)

Data transformations: Real, one-sided HP-filter gap (cyclical component)

Model setup 4/4: The Spillover Indicator

- **Approach:** Weighting early warning indicators (EWI_j) of other countries
- **Sample:** 13 of the 15 EWI-countries
- **Measurement of connectivity:** Foreign claims of the German banking sector towards respective country, as a share of total foreign claims of German banking sector.
- **Calculations for Germany (=i):**

$$S_{i,t} = w_t^{ts} \frac{1}{N-1} \sum_{j \neq i}^N w_{i,j,t}^{cs} FW I_{j,t}$$

$$w_{i,j,t}^{cs} = \frac{\text{Foreign assets in country } j \text{ of banks located in country } i}{\frac{1}{N-1} \sum_{j \neq i}^N \text{Foreign assets in country } j \text{ of banks located in country } i}$$

$$w_{i,t}^{ts} = \frac{\text{Total foreign assets of banks located in country } i}{\text{Total assets (foreign + domestic) of banks located in country } i}$$

Share of i's foreign assets in j, relative to i's total foreign assets (in the sample).

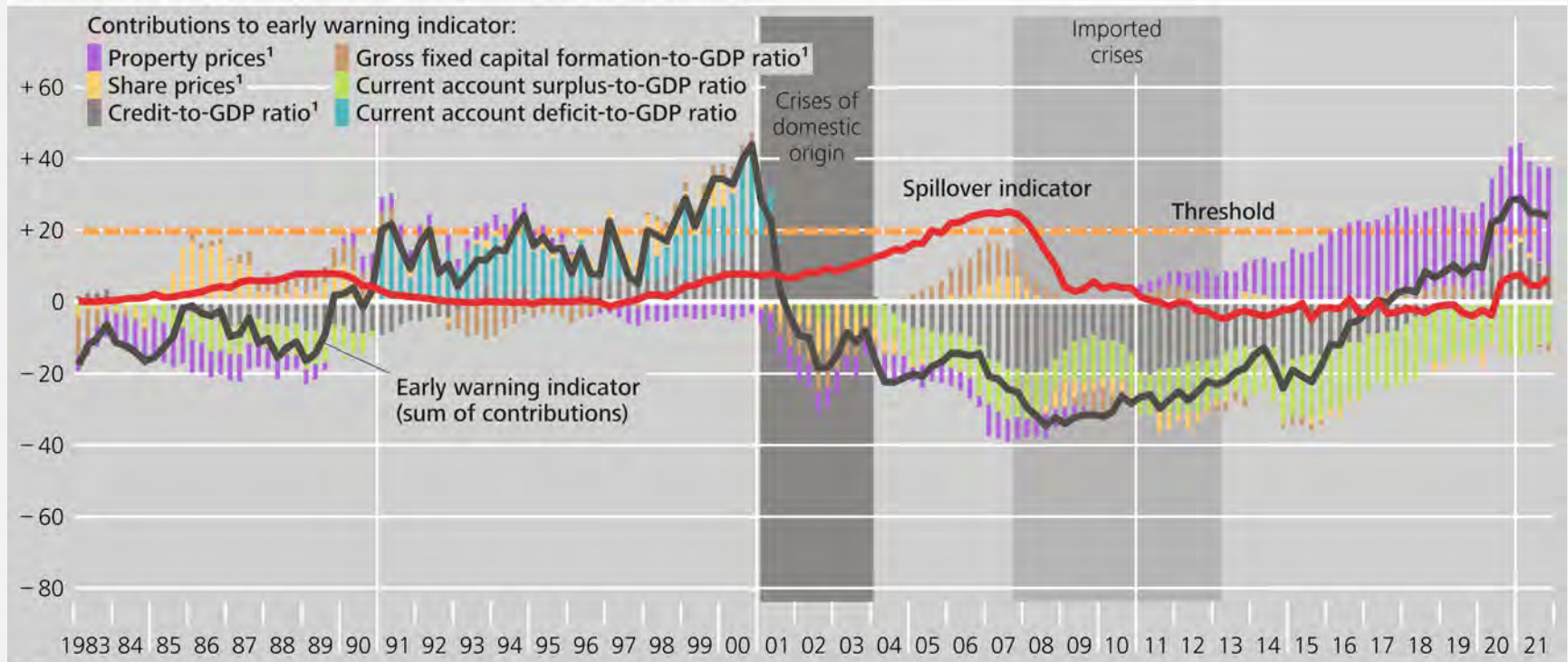
Time varying share of i's foreign claims, relative to i's total (foreign and domestic) claims.

→ Higher weight of countries with larger interconnectedness with the German banking sector.

The Spill-Over-Indicator is elevated during the Global Financial Crises.

Early warning indicator and spillover indicator for Germany

Normalisation: early warning indicator in the United States in Q1 2006 = 100



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Evaluation of the Early Warning Indicator

Model evaluation 1/6: Relative Usefulness requires a mapping of estimated values (pseudo probabilities) into binary signals.

- Threshold is chosen based on **type-1-errors** (missed crises, false negatives) and **type-2-errors** (false alarms, false positive)

	Pre-crisis period	No pre-crisis period
Early-warning signal	True Positive (TP)	False Positive Type-2-error (FP)
No early-warning signal	False Negative Type-1-error (FN)	True Negative (TN)

- Choose optimal threshold:**

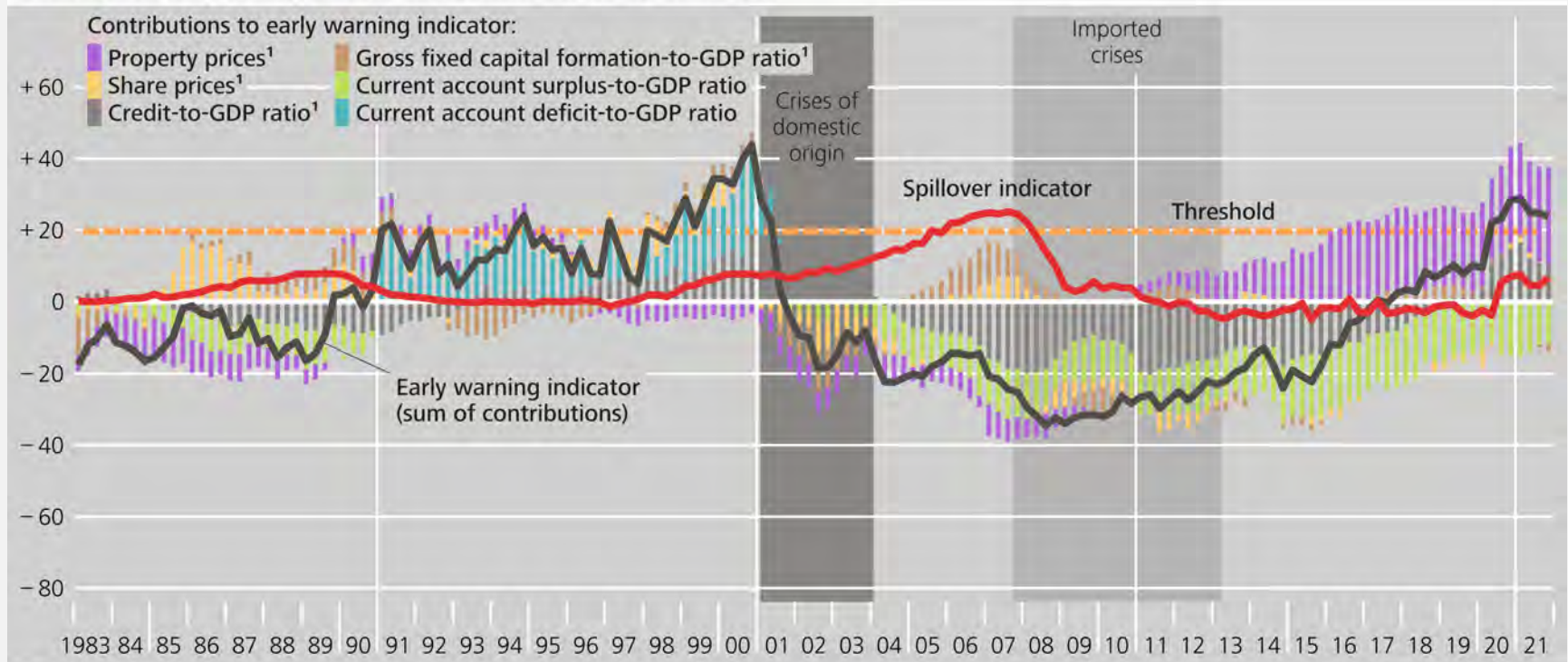
Loss-function:
$$L(\mu) = \mu \frac{FN}{FN+TP} + (1 - \mu) \frac{FP}{FP+TN}$$

Relative usefulness:
$$U_r(\mu) = 1 - \frac{L(\mu)}{\min(\mu, 1-\mu)}$$

Model evaluation 2/6: Relative Usefulness hence requires a threshold.

Early warning indicator and spillover indicator for Germany

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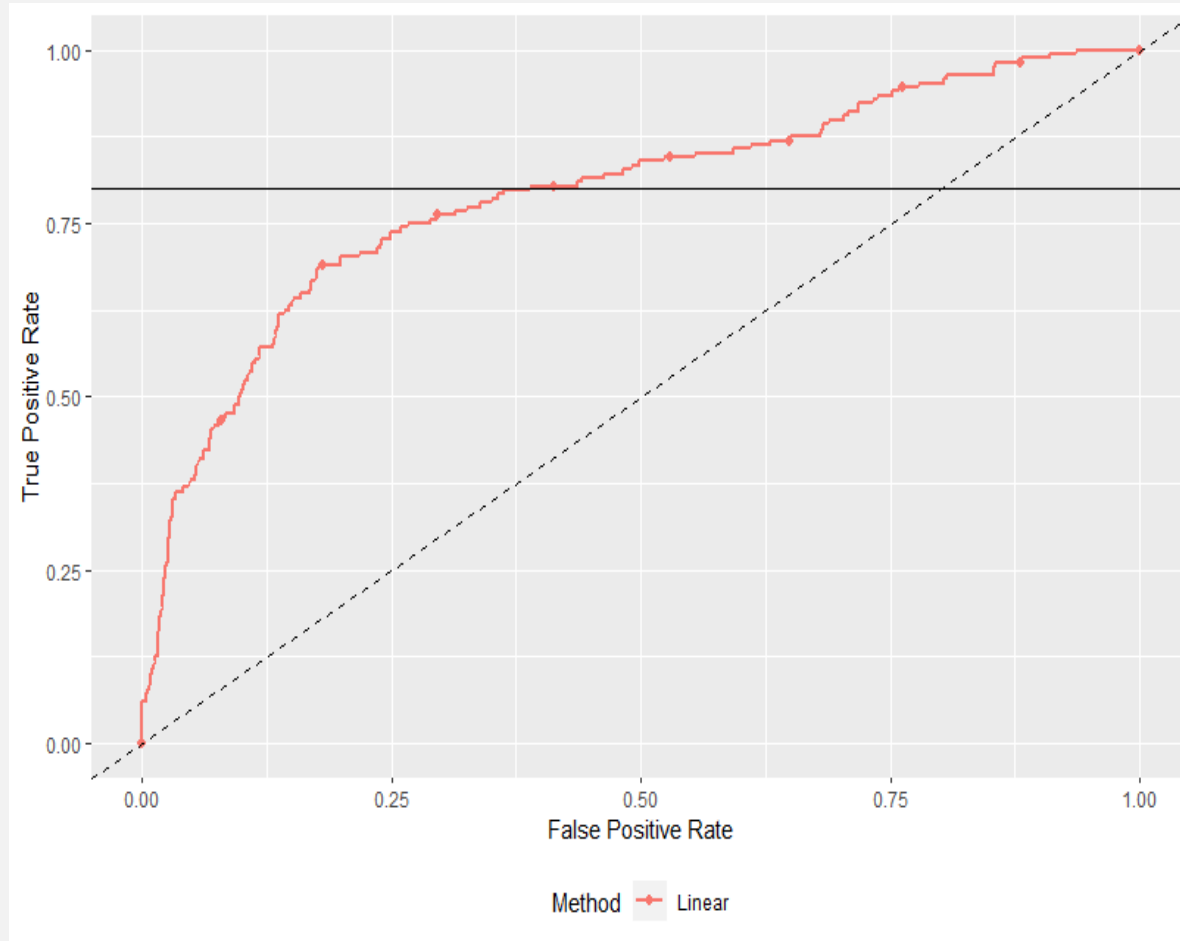
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Model evaluation 3/6: The optimal threshold is chosen using a numerical grid.

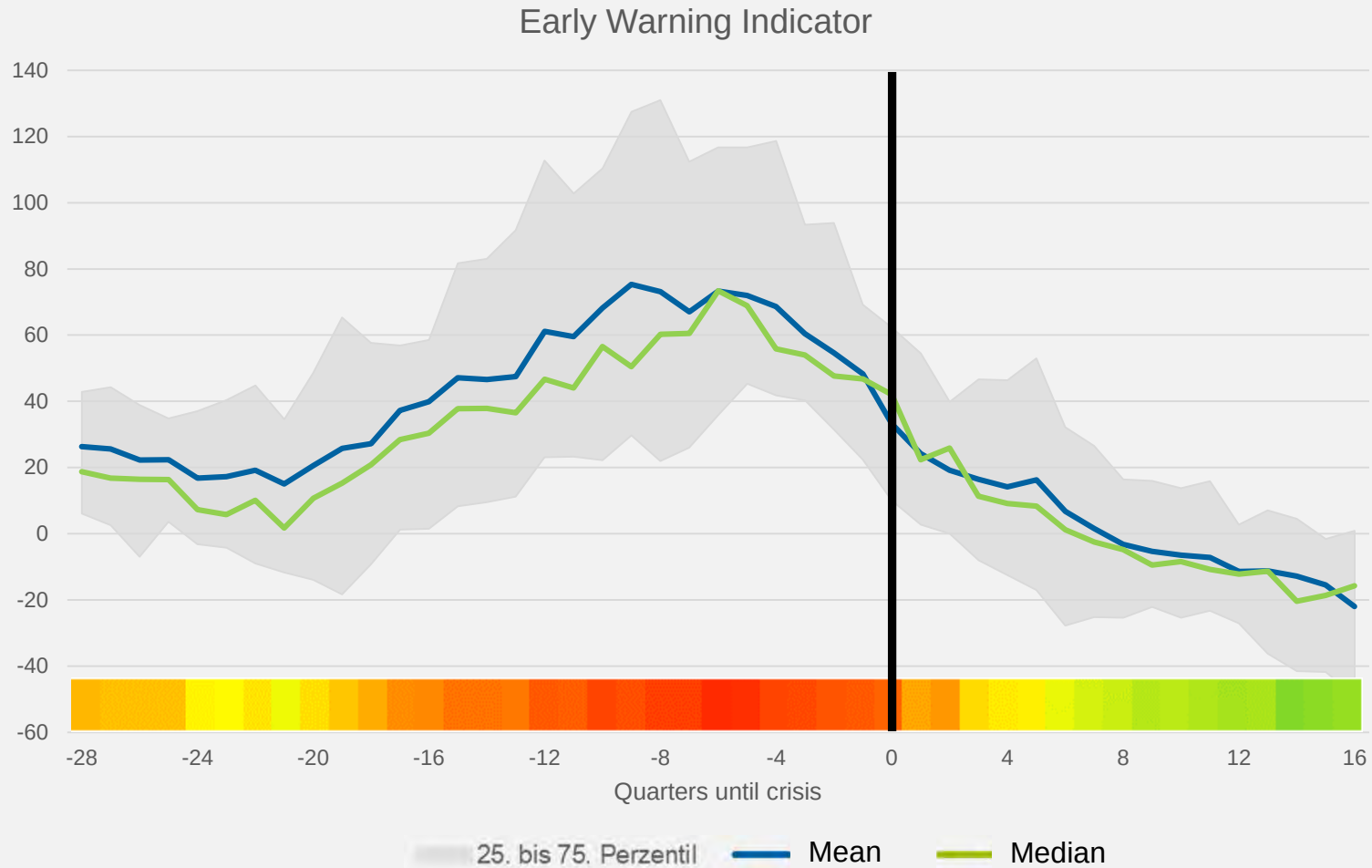
TH_Linear	FPR_Linear	FNR_Linear	U_r_Linear	Helper_Linear
0.08200235	0.3831728	0.2023810	0.4144462	-1.0000000
0.08199937	0.3836797	0.2023810	0.4139394	-1.0000000
0.08199423	0.3841865	0.2023810	0.4134325	-1.0000000
0.08175033	0.3846934	0.2023810	0.4129257	-1.0000000
0.08174272	0.3852002	0.2023810	0.4124188	-1.0000000
0.08173866	0.3857070	0.2023810	0.4119120	-1.0000000
0.08171363	0.3862139	0.2023810	0.4114052	-1.0000000
0.08167512	0.3867207	0.2023810	0.4108983	-1.0000000
0.08161513	0.3872276	0.2023810	0.4103915	-1.0000000
0.08159089	0.3877344	0.2023810	0.4098846	-1.0000000
0.08152342	0.3882413	0.2023810	0.4093778	-1.0000000
0.08142872	0.3887481	0.2023810	0.4088709	-1.0000000
0.08135202	0.3887481	0.1964286	0.4148233	0.4148233
0.08127895	0.3892549	0.1964286	0.4143165	0.4143165
0.08111189	0.3897618	0.1964286	0.4138096	0.4138096
0.08103296	0.3902686	0.1964286	0.4133028	0.4133028
0.08089878	0.3907755	0.1964286	0.4127960	0.4127960
0.08089812	0.3912823	0.1964286	0.4122891	0.4122891

Model evaluation 4/6: Area Under the Curve does not depend on a specific threshold nor on a preference parameter.

- AUC
 - 1 for perfect forecast
 - >0.5 for informative forecast
 - 0.5 for naive forecast
 - <0.5 for misleading forecast



The Early Warning Indicators typically rise 5-12 quarters prior to a financial crisis.

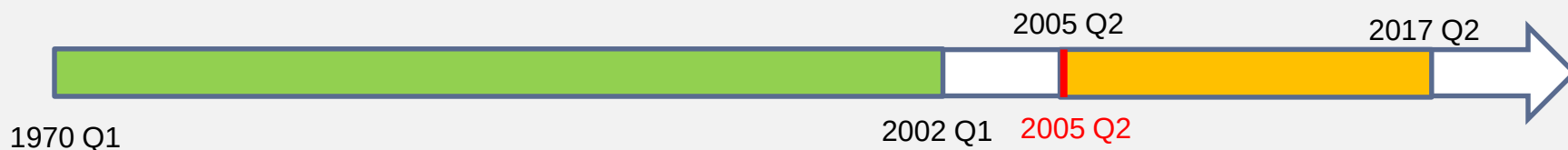


Model evaluation 5/6: In-Sample and Out-of-Sample Evaluation

- In-Sample-Estimation
 - Prediction of all observations in training sample and compute relative usefulness/AUC. (All available data enters the training sample=estimation sample).
- Out-of-Sample-Estimation
 - Quasi Real-Time Exercise
 - Recursively estimate the model and evaluate the model's performance for every quarter with all information that would have been available in real time at this quarter.

Model evaluation 6/6: Out-of-Sample Evaluation

- **Training sample: 1970 Q1 – 2002 Q1; Out-of-Sample-Window: 2005 Q2 – 2017 Q2**
- First iteration: Estimate value for 2005 Q2 based only on information available in this quarter (coefficients as of 2002 Q1, explanatory variables as of 2005 Q2)



- Second iteration: Estimate value for 2005 Q3 based on all information available in this quarter



- Last iteration:



Variable selection of the Early Warning Indicator

Variable selection 1/9: Sources of vulnerabilities and corresponding indicators in the literature (cited in Beutel et. al 2019).

- Asset Prices

- “Banking crises have often been preceded by asset price booms (Reinhart & Rogoff, 2008)”

- Credit Developments

- “High private sector indebtedness poses risks to the financial system when asset price booms are debt-financed, asset prices decrease and borrowers are unable to repay their debt (Kindleberger and Aliber, 2005; Jordà et al., 2015)”

- Macroeconomic Environment

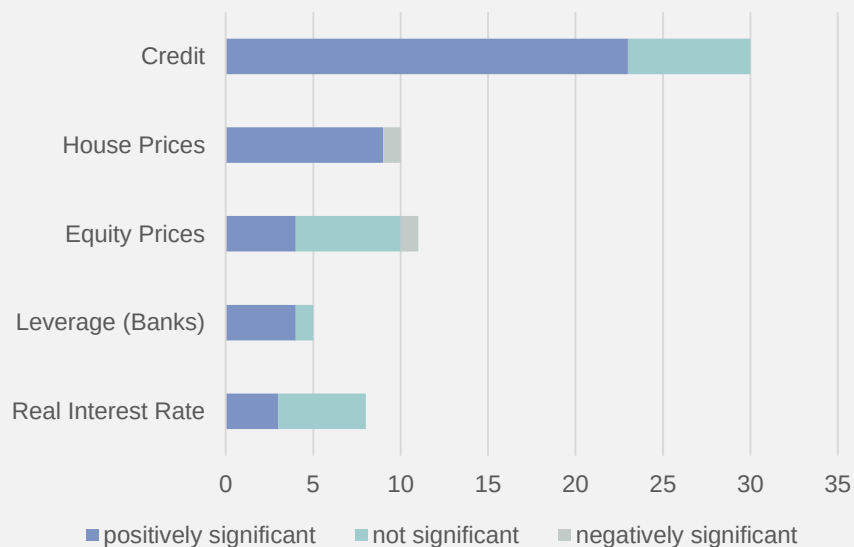
- “Real economic downturns may lead to repayment difficulties on the borrower side inducing asset price declines and financial sector difficulties (Allen and Gale, 2007)”

- External and Global Imbalances

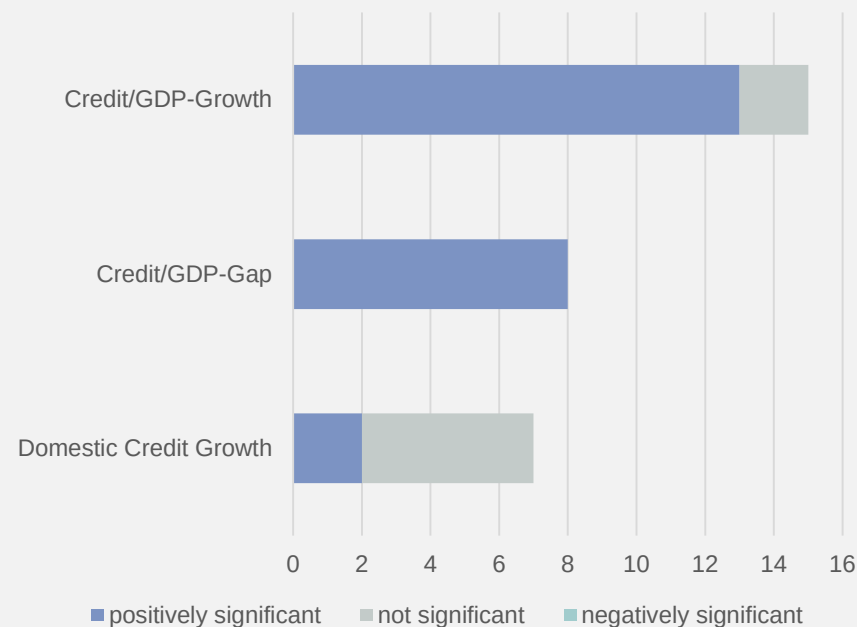
- “Large capital inflows from abroad may support asset price booms and induce a reversal in asset prices when these inflows decline or stop (Kaminsky and Reinhart, 1999; Calvo, 1998)”

Variable selection 2/9: The literature identifies several early warning indicators (Meta analysis).

Significance of Early Warning Indicators for Banking Crises
Financial Indicators

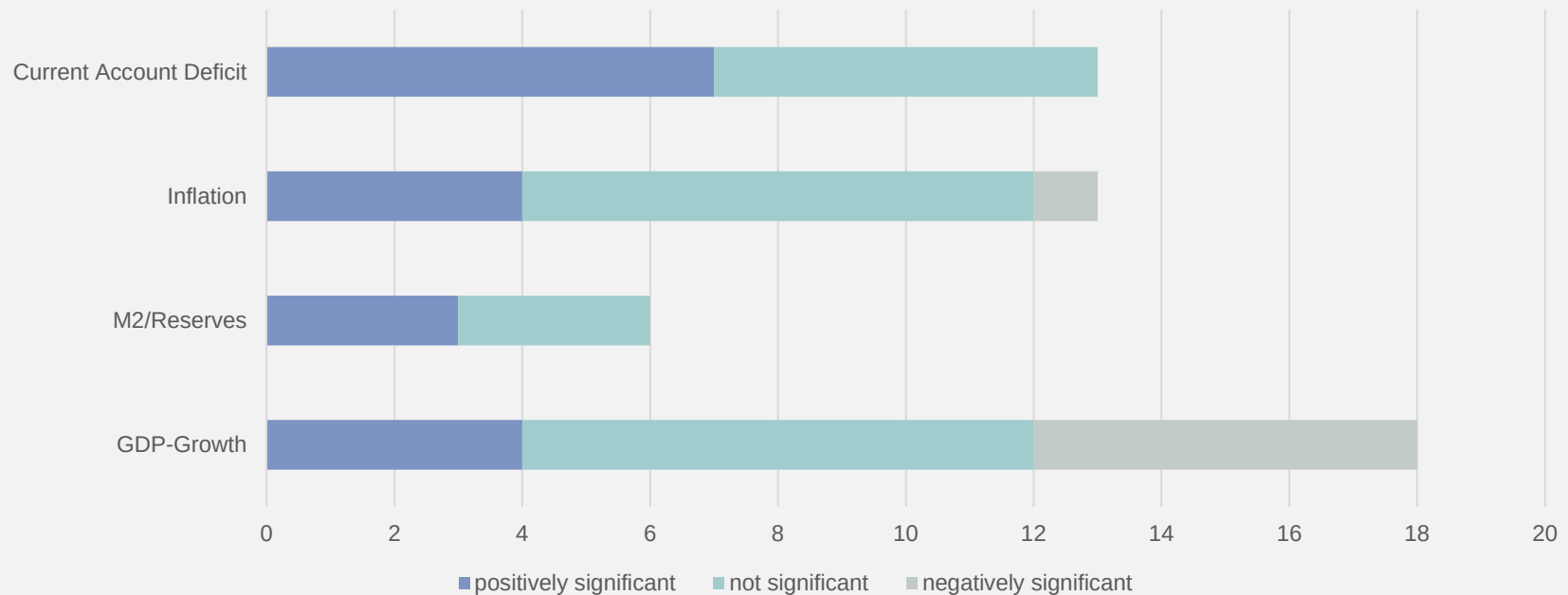


Different Credit Measures

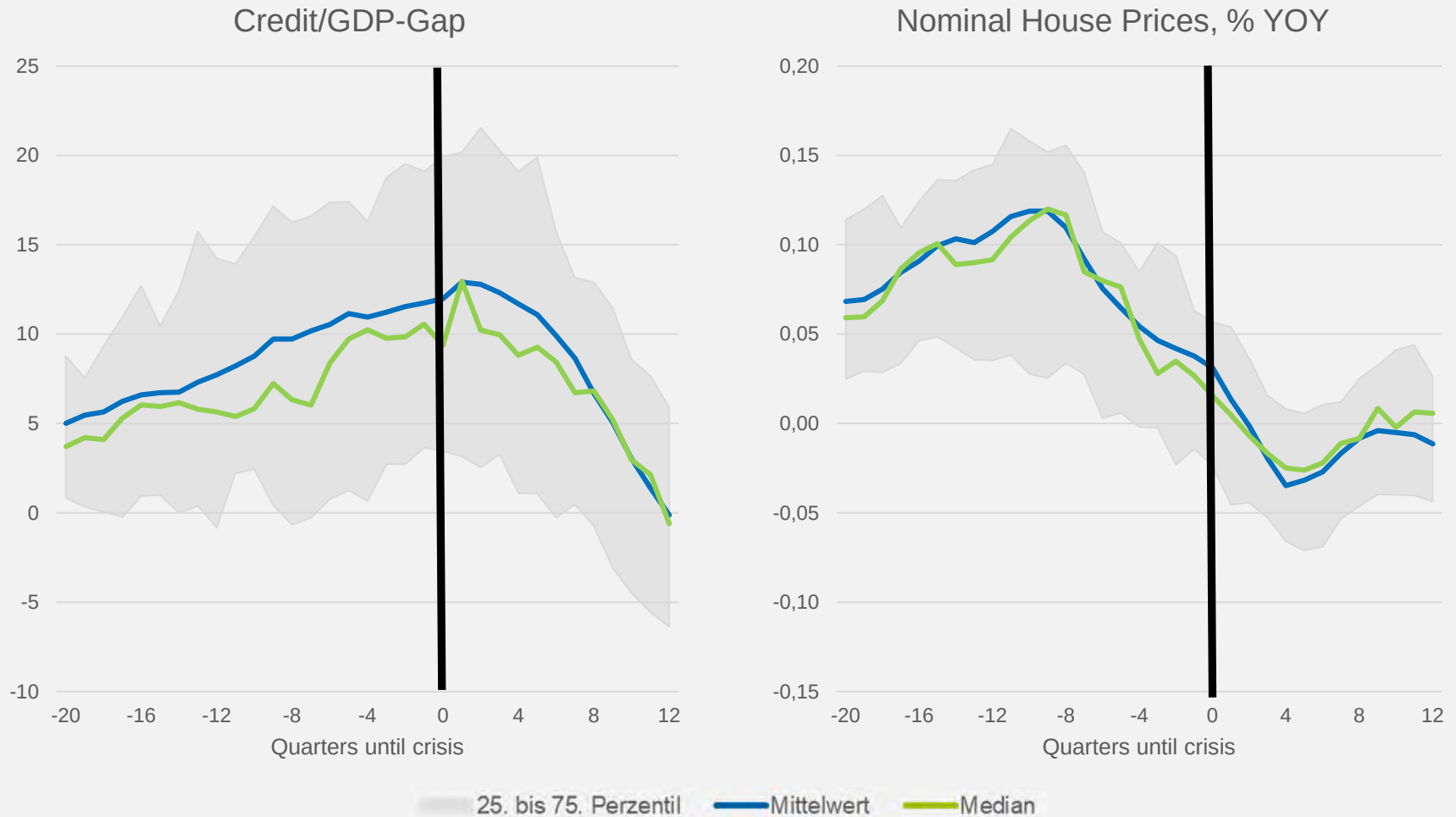


Variable selection 3/9: Macro Environment and external indicators seem to be less significant indicators (Meta analysis).

Significance of Early Warning Indicators for Banking Crises
Macro Environment and External Indicators



Variable selection 4/9: Real estate prices appear to give early and reliable warning signals (Event plots).



Variable selection 5/9: Performance Evaluation of different sets of explanatory variables in Beutel et. al. 2019.

Table B6
Variables used in specifications (1)–(4).

Credit and asset prices (1)	Macro (2)	External (3)	All (4)
Total credit-to-GDP gap Real residential real estate price gap Real equity price gap	Real GDP gap Inflation rate 3-month real money market rate Gross fixed capital formation-to-GDP gap	Current account-to-GDP ratio Real effective exchange rate gap Real oil price gap	Total credit-to-GDP gap Real residential real estate price gap Real equity price gap Real GDP gap Inflation rate 3-month real money market rate Gross fixed capital formation-to-GDP gap Current account-to-GDP ratio Real effective exchange rate gap Real oil price gap

Variable selection 6/9: Performance Evaluation of different sets of explanatory variables in Beutel et al. 2019.

Table 1
In-sample relative usefulness.

	(1) Credit/asset prices	(2) Macro	(3) External	(4) All
logit	0.347** [0.185, 0.440]	0.202*** [0.018, 0.236]	0.390* [0.243, 0.422]	0.511* [0.208, 0.561]
trees	0.432** [0.158, 0.601]	0.000*** [−0.031, 0.216]	0.000** [0.000, 0.410]	0.674* [0.230, 0.711]
knn	0.693* [0.353, 0.693]	0.965 [0.265, 0.965]	0.685 [0.302, 0.685]	0.955 [0.370, 0.955]
rf	0.966 [0.411, 0.966]	0.966 [0.240, 0.966]	1.000 [0.299, 1.000]	0.992 [0.347, 0.992]
svm	0.362** [−0.006, 0.483]	0.481** [0.129, 0.489]	0.000** [−0.258, 0.320]	0.784 [0.334, 0.784]
nen	0.456** [0.201, 0.503]	0.266*** [0.042, 0.294]	0.377* [0.263, 0.419]	0.764 [0.338, 0.793]

Note: Highest usefulness on each dataset in bold. Stars indicate whether the respective usefulness is significantly below the best performance on the same dataset. Numbers in brackets indicate 90% confidence bands.

* For the 10% level.

** For the 5% level.

*** For the 1% level.

Variable selection 7/9: Performance Evaluation of different sets of explanatory variables in Beutel et al. 2019.

Table 2

Out-of-sample relative usefulness.

	(1) Credit/asset prices	(2) Macro	(3) External	(4) All
logit	0.368 [0.143, 0.431]	−0.236 [−0.386, −0.036]	0.438 [0.309, 0.563]	0.605 [0.222, 0.605]
trees	0.201 [0.084, 0.317]	−0.605 [−0.605, −0.087]	0.390*** [0.072, 0.390]	0.126*** [−0.037, 0.313]
knn	0.247 [0.088, 0.384]	−0.087 [−0.137, 0.034]	0.293*** [0.133, 0.417]	−0.062*** [−0.166, 0.042]
rf	0.246 [0.129, 0.350]	−0.245 [−0.323, −0.132]	0.138*** [0.055, 0.283]	−0.003*** [−0.141, 0.146]
svm	0.218* [0.018, 0.326]	− 0.015 [−0.257, 0.060]	0.202** [0.043, 0.409]	−0.065*** [−0.173, 0.130]
nen	0.251 [0.080, 0.451]	−0.349 [−0.382, −0.048]	0.434 [0.322, 0.551]	0.092** [−0.041, 0.250]

Note: Highest usefulness on each dataset in bold. Stars indicate if the respective usefulness is significantly below the best performance on the same dataset.

Numbers in brackets indicate 90% confidence bands.

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** For the 5% level.

*** For the 1% level.

Variable selection 8/9: Estimation results of the linear model

```
GAP400_CT2GDP      0.003***  
                    (0.0005)  
  
GAP400_RREPR_rel   0.284***  
                    (0.044)  
  
GAP400_EQPRR_rel   0.039**  
                    (0.016)  
  
GAP1.6_GFCF2GDP    0.023***  
                    (0.006)  
  
negCA2GDP          -0.024***  
                    (0.003)  
  
posCA2GDP          -0.004**  
                    (0.002)  
  
Constant           0.043***  
                    (0.008)
```

=====

Note: *p<0.1; **p<0.05; ***p<0.01

Models	AUC_IS	U_r_IS	U_r_OOS
Linear	0.7920981	0.4148233	0.5422029
Linear_Credit	0.7534634	0.4143165	0.4681503

Variable selection 9/9: Variable Selection for ECB's d-SRI (Lang et al. 2019).

Selection of sub-indicators for the d-SRI

Five indicator categories are included in the d-SRI that cover the key categories recommended by the ESRB for monitoring cyclical systemic risks.²³ Basel Committee on Banking Supervision (BCBS) guidance, the Capital Requirements Directive (CRD IV) and ESRB Recommendation ESRB/2014/1 assign a central role to the credit-to-GDP gap for measuring cyclical systemic risks. However, additional indicator categories can cover other dimensions of cyclical systemic risks and complement signals from the Basel gap. The d-SRI is therefore designed to cover five of the indicator categories for cyclical systemic risks that are recommended by the ESRB (see ESRB/2014/1):²⁴

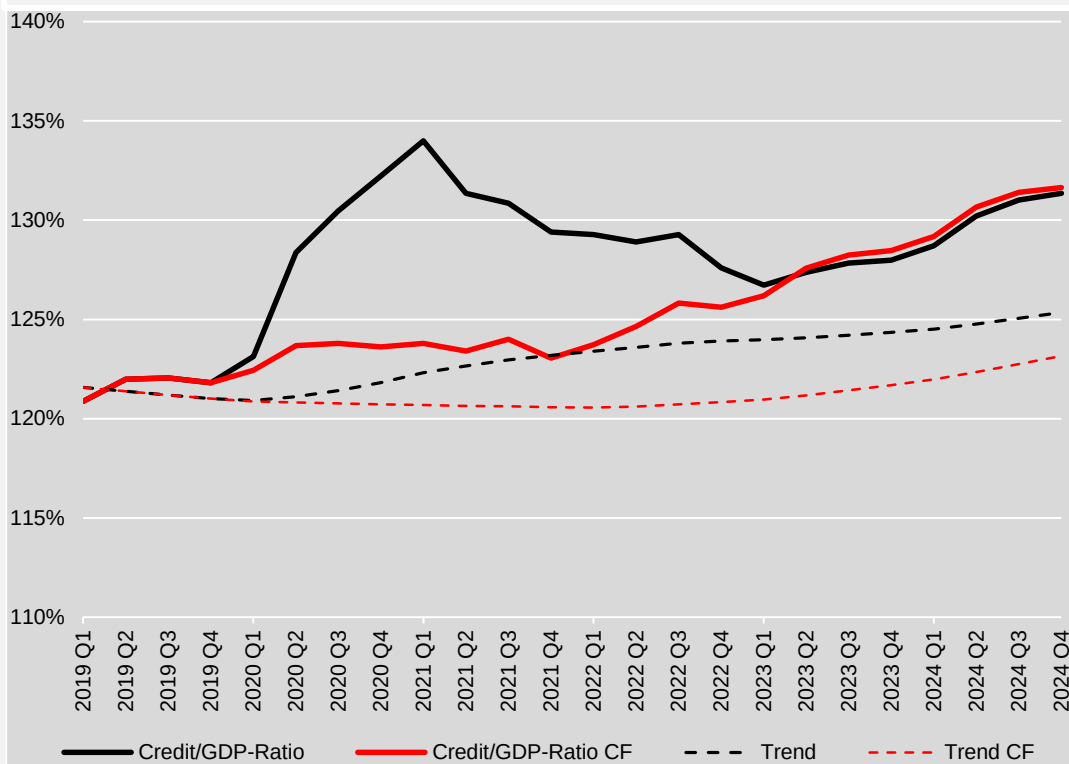
- (i) measures of potential overvaluation of property prices;
- (ii) measures of credit developments;
- (iii) measures of external imbalances;
- (iv) measures of private sector debt burden;
- (v) measures of potential mispricing of risk.

- The d-SRI uses five indicator categories.
- In each category, the best performing indicator from single variable regressions is chosen.

EWI counterfactual analysis and nowcasting

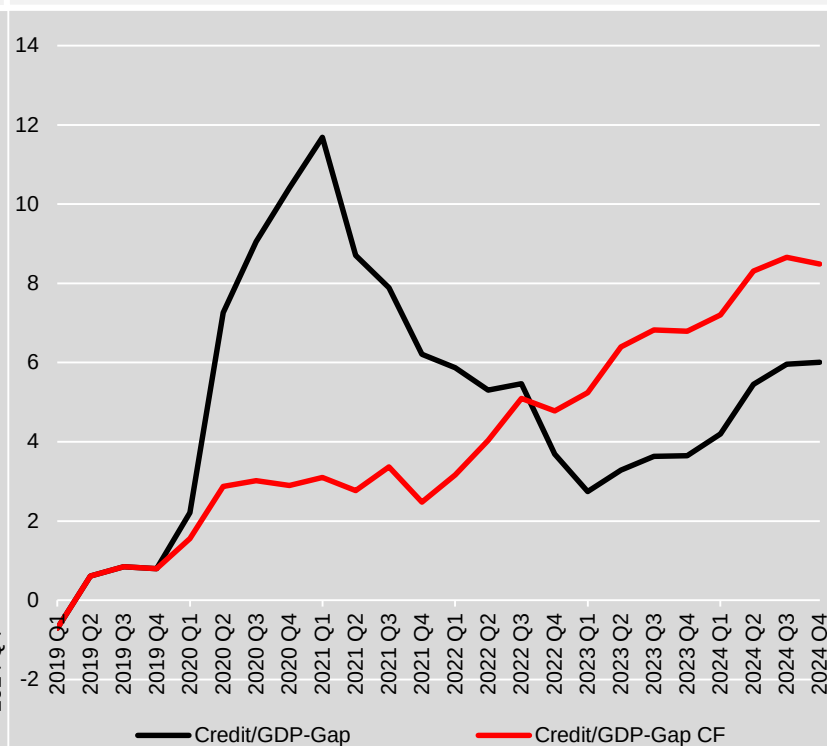
GDP developments in times of Covid influence risk indicators during and after the crisis.

Credit/GDP-Ratio – Counterfactual Analysis



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Credit/GDP Gap – Counterfactual Analysis



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Background Material / Links:

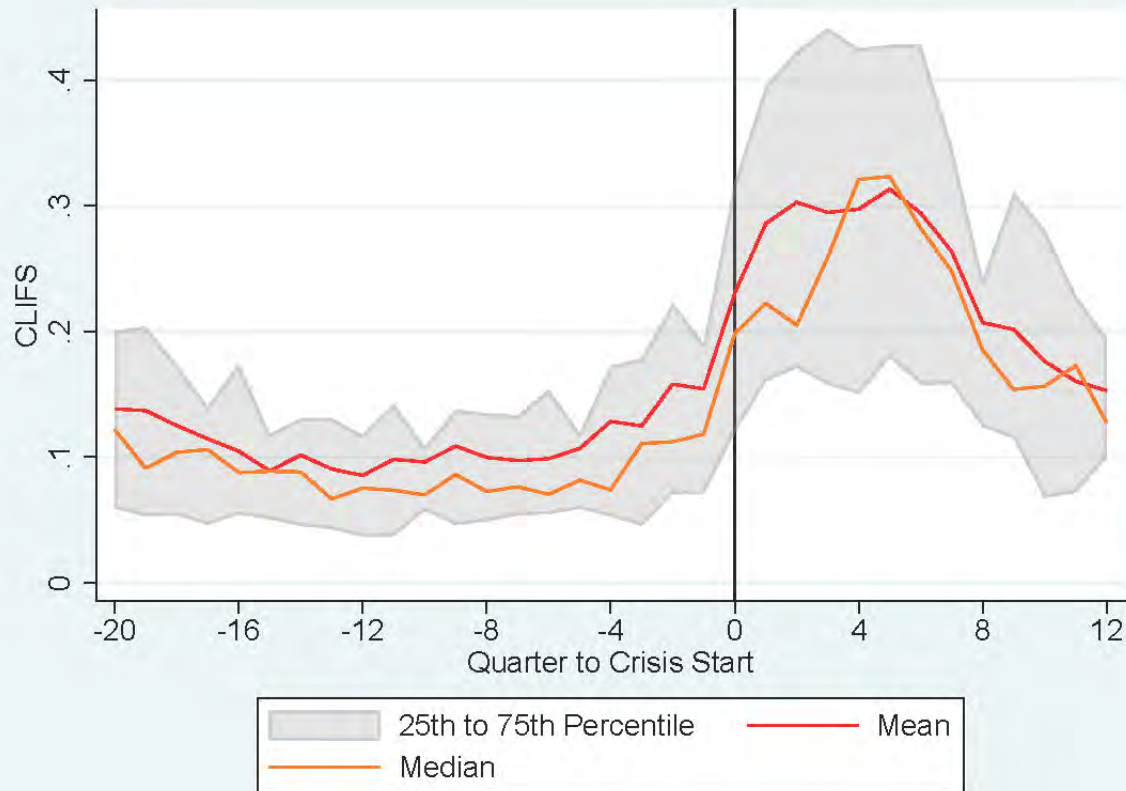
- **Johannes Beutel, Sophia List, Gregor von Schweinitz:**
Does machine learning help us predict banking crises?,
Journal of Financial Stability, Volume 45, 2019
<https://www.sciencedirect.com/science/article/pii/S1572308918305801?via%3Dihub>
- **Jan Hannes Lang, Cosimo Izzo, Stephan Fahr, Josef Ruzicka:**
Anticipating the bust: a new cyclical systemic risk indicator to assess the likelihood and severity of financial crises,
ECB Occasional Paper No. 219 (2019)
[Anticipating the bust: a new cyclical systemic risk indicator to assess the likelihood and severity of financial crises \(ssrn.com\)](https://ssrn.com/abstract=3411111)
- **Bundesbank: Financial Stability Reviews**
2019: p. 39-41
2018: p. 47-49
2017: p. 45-48
[Financial Stability Review | Deutsche Bundesbank](https://www.bundesbank.de/pressroom/financial-stability-reviews)

Questions?

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- Variable selection of the EWI
 - Literature/Meta-Analysis
 - Eventplots
 - EWI Model evaluation
- Counterfactual EWI analysis in light of the Corona pandemic

Annex

The ECB's CLIFS (Country level indices of financial stress) begin to rise only shortly before crises.



Variable selection 10/10: Variable Evaluation for ECB's d-SRI (Lang et al. 2019).

Best indicator for each variable category based on a weighted average of in-sample and out-of-sample performance

Indicator	Final Ranking [Weighted average ranking (1/3 to out-of-sample and 2/3 to in-sample)]	In-sample					Out-of-sample	
		Rank based on weighted average (50%-35%-15%)	Weighted average (rel. weights 50%-35%-15%)	12-5 check AUROC (rel. weight of 15%)	12-5 bench AUROC (rel. weight of 50%)	16-5 bench AUROC (rel. weight of 35%)	Rank by relative usefulness	Relative Usefulness
I – Bank credit to NFFS as % of GDP, 2-year change	1 [3]	1	0.83	0.85	0.84	0.82	7	0.38
II – Total credit to HHs as % of GDP, 2-year change	3 [5.7]	7	0.80	0.82	0.79	0.79	3	0.42
III – Debt service ratio, 2-year change	8 [11.7]	11	0.77	0.76	0.79	0.74	13	0.34
IV – Monetary aggregate M3, 3-year real growth	12 [15.3]	21	0.74	0.75	0.75	0.72	4	0.41
V – Total consolidated credit as % of GDP, 1-year change	15 [24.7]	15	0.76	0.79	0.76	0.74	44	0.25
VI – Total consolidated credit to NFCs, 1-year real growth	18 [29.3]	22	0.74	0.77	0.74	0.72	44	0.25
VII – House price-to-income ratio, 3-year change	27 [36.3]	24	0.73	0.76	0.74			
VIII – Current account balance as % of GDP	41 [45.7]	54	0.70	0.68	0.71			
IX – Equity price, 3-year real growth	56 [58]	86	0.65	0.68	0.65			
X – Basel gap: Total credit as % of GDP, GAP (400'000)	59 [58.3]	46	0.71	0.73	0.72			

- Evaluation based on weighted average between in-sample and out-of-sample performance.
- Relative Usefulness and AUC use as evaluation criteria.

- (i) the two-year change in the bank credit-to-GDP ratio;
- (ii) the two-year growth rate of real total credit;
- (iii) the two-year change in the debt-service-ratio;
- (iv) the three-year change in the RRE price-to-income ratio;
- (v) the three-year growth rate of real equity prices;
- (vi) the current account-to-GDP ratio.